

# CAT 2025

MBA FASTRACK BATCH

*percentages.*

Lecture - 02

Data Interpretation

Pie Charts & Caselets

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# TOPICS

*to be covered*

19-20 Q.

1

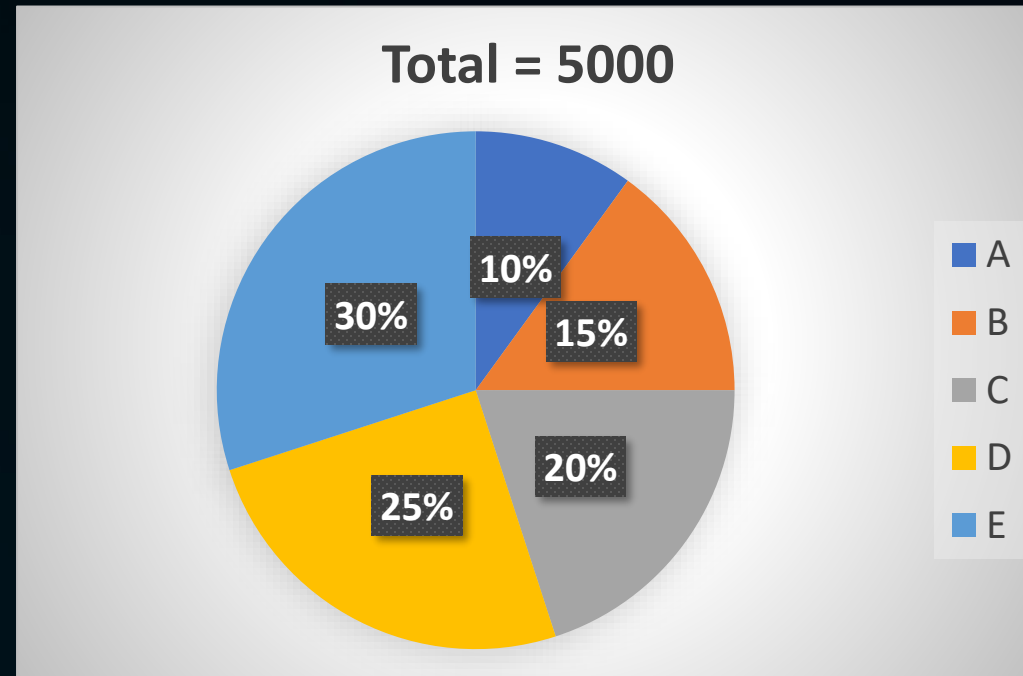
DI Sets based on Pie Charts

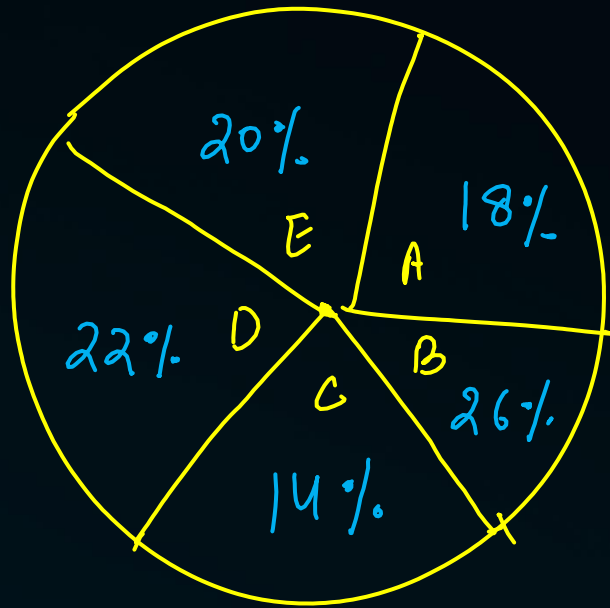
2

DI Sets based Caselets



# PIE CHARTS





A =

B =

C =

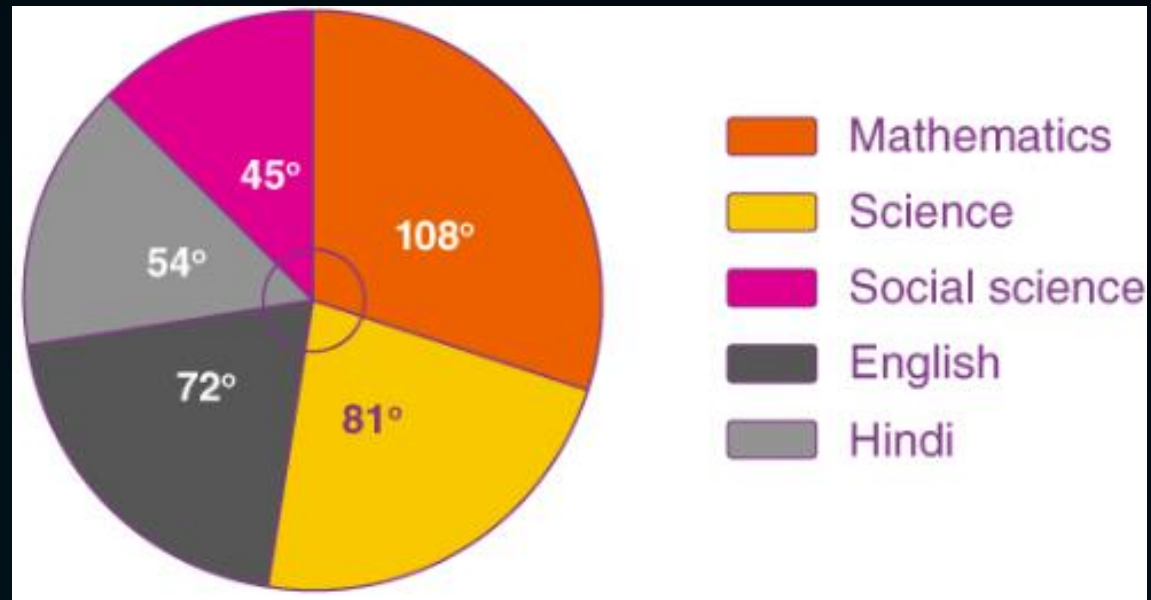
D =

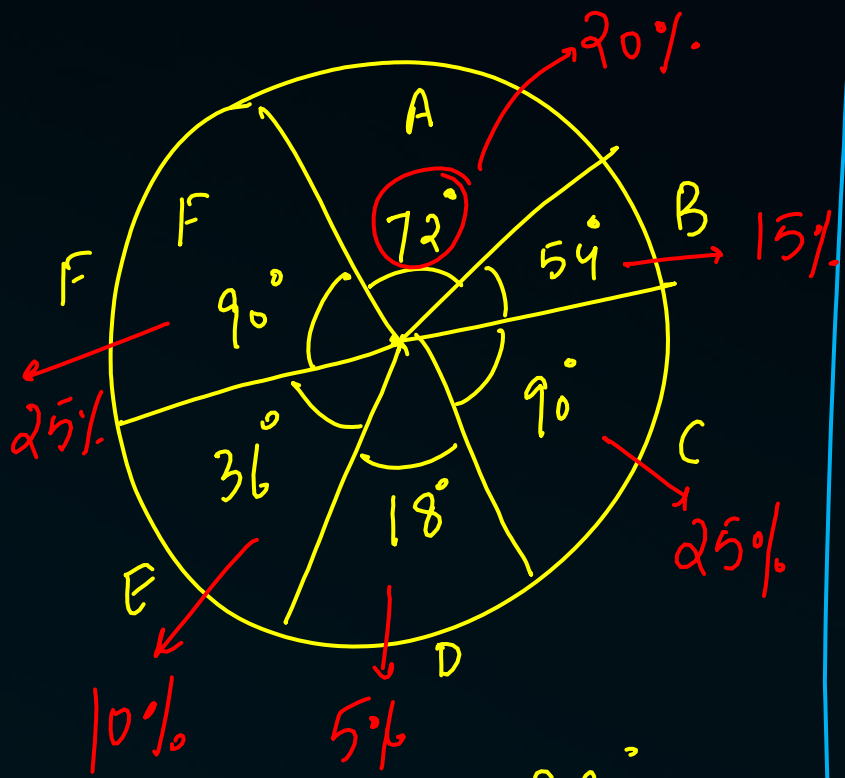
E =

Year 2020

percentage Total = 100%

Total Popn of 5 countries = 250 million.





Total Angle =  $360^\circ$

Total popu = 250 mil

$$A = \frac{72^\circ}{360^\circ} \times 250 = 50 \text{ mil.}$$

B =

$$\text{Total} = 360^\circ$$

$$10\% \rightarrow 36^\circ$$

$$5\% \rightarrow 18^\circ$$

$$2.5\% \rightarrow 9^\circ$$

$$36 \times 2 = 72^\circ \rightarrow 20\%$$



# SET - 1

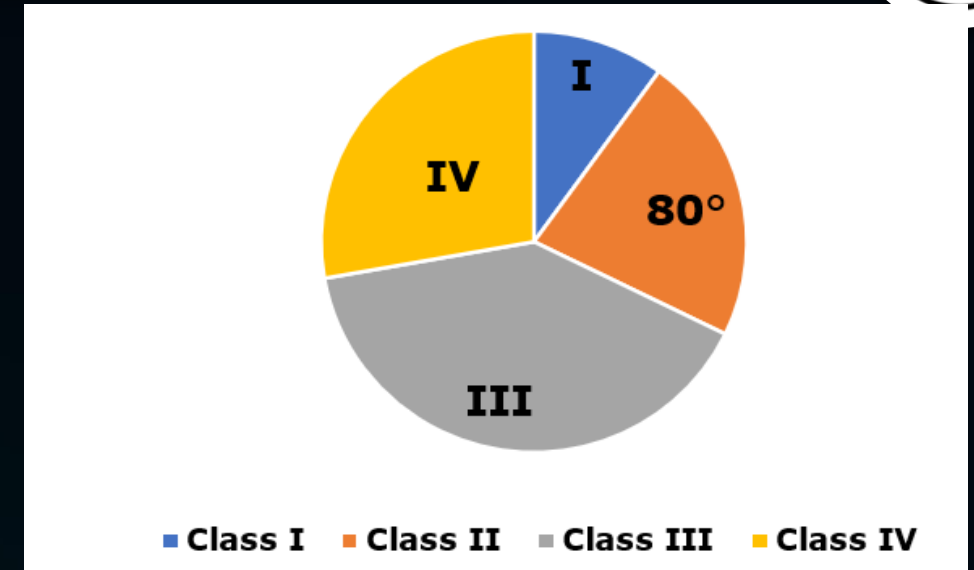


Read the following data and answer the given questions.



XYZ school has four primary classes: I, II, III, and IV. Each of these classes have a certain number of students. (both boys and girls).

The pie-chart given below shows the degree-distribution of boys in 4 classes. Some of the data is the missing.



It is also known that:

- The ratio of the values of degrees of Class I to IV in the pie chart is 9:25.
- The number of boys in Class III is 960 more than Class II.
- The number of boys in Class IV is  $(5/4)$ th of the number of boys in Class II.
- 60% of the students in the school are boys.

## QUESTION- 1



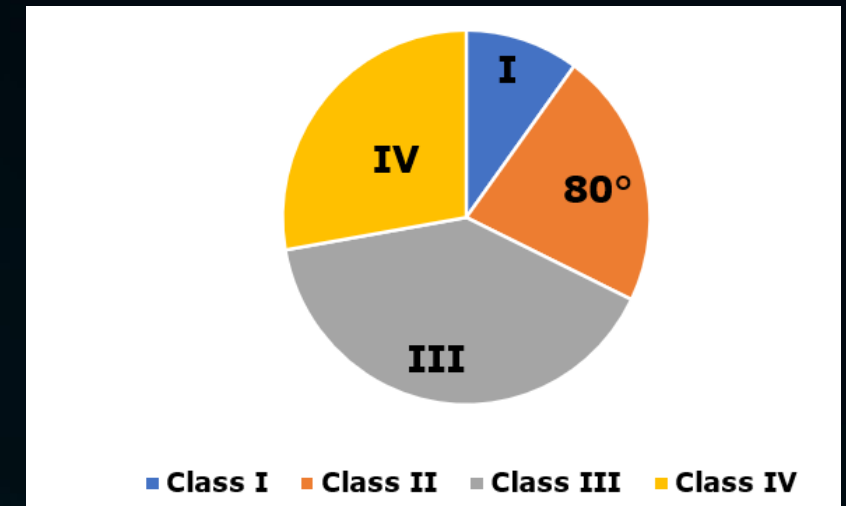
#Q. What is the total number of students in XYZ school?

A. 5,400

B. 7,200

☒ C. 9,000

D. 10,800



- a. The ratio of the values of degrees of Class I to IV in the pie chart is 9:25.
- b. The number of boys in Class III is 960 more than Class II.
- c. The number of boys in Class IV is  $(5/4)$ th of the number of boys in Class II.
- d. 60% of the students in the school are boys.

## QUESTION- 2



#Q. If the ratio of the number of girls in Class I to the number of boys in Class II is 3:5, what is the difference between the number of boys in Class IV and the number of girls in Class I?

A. 700

B. 780

C. 880

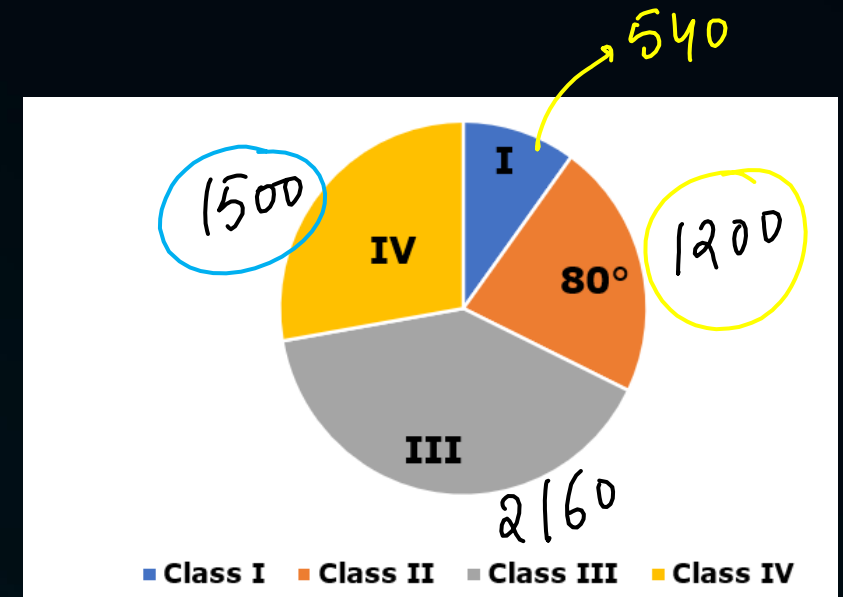
D. 960

$$\frac{G1}{B2} = \frac{3}{5} = \frac{G1}{1200}$$

$$G1 = 720$$

$$B4 = 1500$$

$$\rightarrow 780$$



- The ratio of the values of degrees of Class I to IV in the pie chart is 9:25.
- The number of boys in Class III is 960 more than Class II.
- The number of boys in Class IV is (5/4)th of the number of boys in Class II.
- 60% of the students in the school are boys.

## QUESTION- 3



#Q. If the number of students in Class II is 20% more than Class I, number of students in Class III is  $\frac{1}{3}$  33.33% more than Class II and number of students in Class IV is 900 more than Class III, how many girls are there in Class III?

A. 450

B. 320

C. 240 ✓

~~D. The given condition is not possible~~

①  $5n = 1500$

②  $6n = 1800$

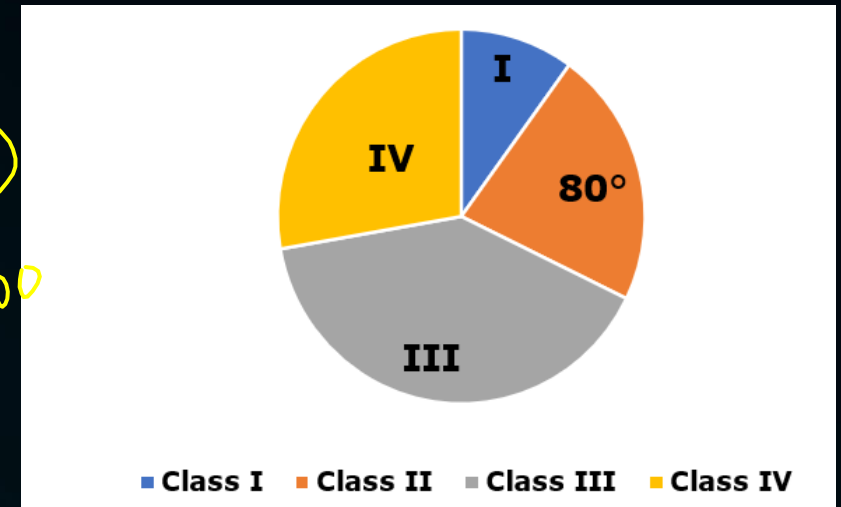
③  $8n = 2400$

④  $8n + 900 = 3300$

9000

$n = 300$

$2400 - 2160 = 240$



- The ratio of the values of degrees of Class I to IV in the pie chart is 9:25.
- The number of boys in Class III is 960 more than Class II.
- The number of boys in Class IV is  $(5/4)$ th of the number of boys in Class II.
- 60% of the students in the school are boys.

## QUESTION- 4



#Q. If the number of students in Class III is 2,880, then what percentage of the total number of girls in the school comprises the number of girls in Class III?

A. 10%

B. 15%

C. 20%

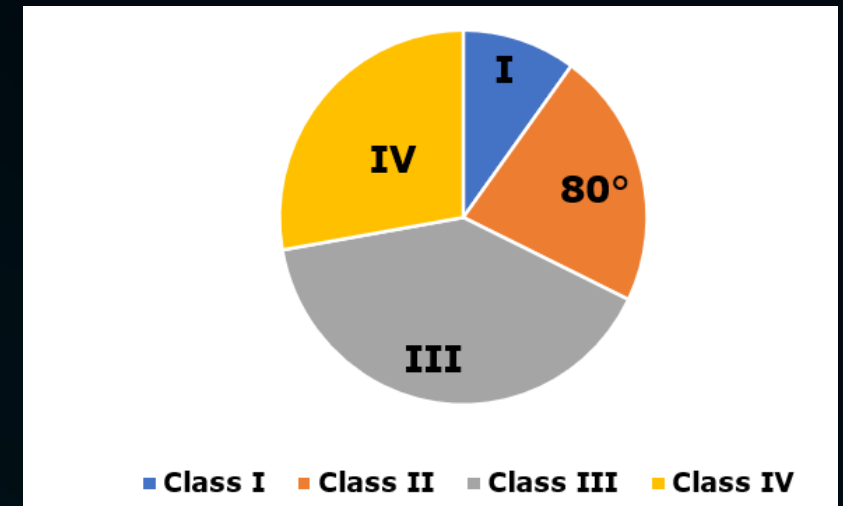
D. 25%

Handwritten calculations:

$$\begin{array}{l} 2880 \\ \swarrow \\ \text{Boys} = 2160 \end{array}$$
$$\begin{array}{l} \searrow \\ \text{Girls} = 720 \end{array}$$

$$\frac{720}{3600}$$

3600



- The ratio of the values of degrees of Class I to IV in the pie chart is 9:25.
- The number of boys in Class III is 960 more than Class II.
- The number of boys in Class IV is  $(5/4)$ th of the number of boys in Class II.
- 60% of the students in the school are boys.

## Analysis



Boys:

$$1 \rightarrow 9n \rightarrow 36^\circ = 540$$

$$2 \rightarrow 8^\circ \rightarrow 1200$$

$$3 \rightarrow 144^\circ \rightarrow 2160$$

$$4 \rightarrow 25n = \frac{5}{4}(8^\circ) = 10^\circ \rightarrow 1500$$

5400

Students = 9000

Girls = 3600

$$144^\circ - 8^\circ = 64^\circ$$

$$64^\circ = 960 \text{ Boys}$$

$$1^\circ = 15 \text{ Boys}$$

80°

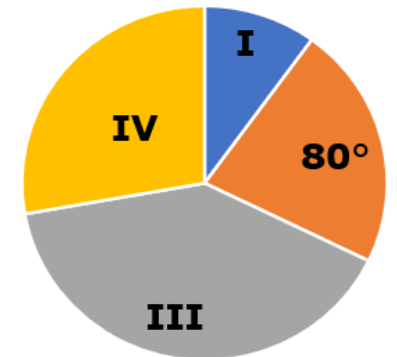
The pie-chart given below shows the degree-distribution of boys in 4 classes.

a. The ratio of the values of degrees of Class I to IV in the pie chart is 9:25.

b. The number of boys in Class III is 960 more than Class II.

c. The number of boys in Class IV is  $(5/4)$ th of the number of boys in Class II.

d. 60% of the students in the school are boys.



■ Class I ■ Class II ■ Class III ■ Class IV

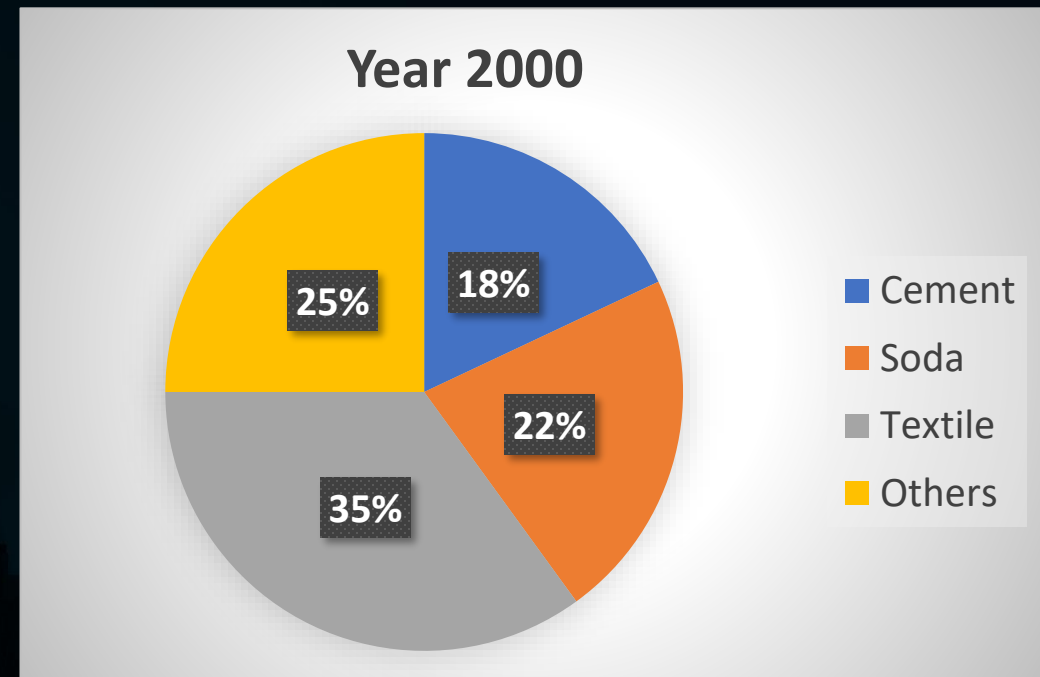
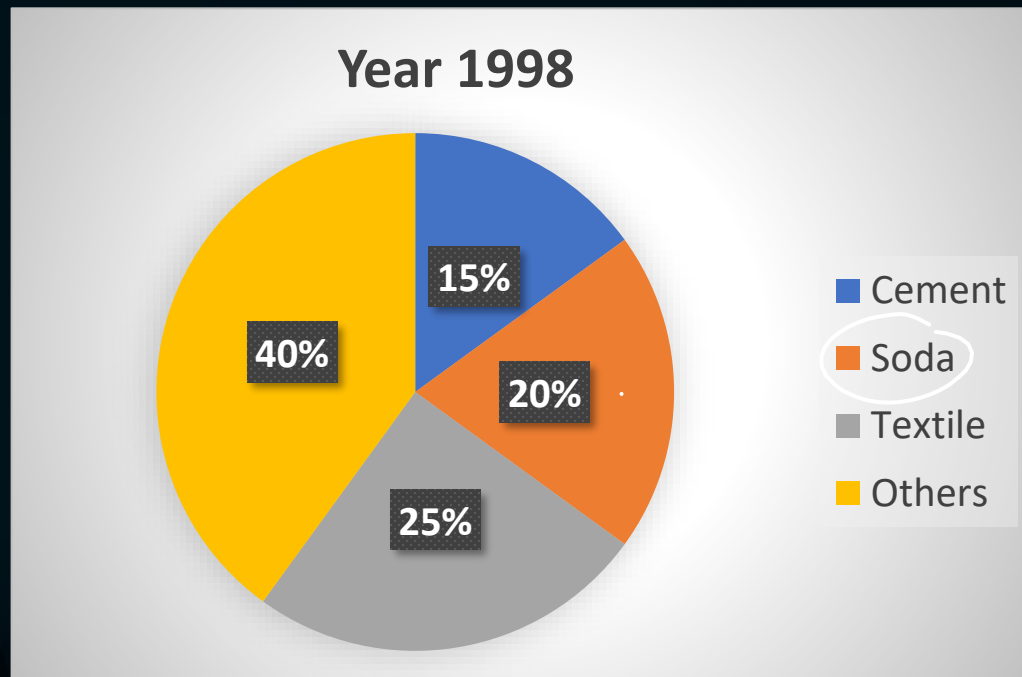


**SET - 2**

Read the following data and answer the given questions.



The percentage of revenue that comes from four different product segments of a manufacturing company are shown in the following pie charts. The figures pertaining to the year 1998 and 2000 are given in the right side. The total sales in the year 1998 were Rs. 120 crores. It was Rs. 150 crores in 2000.





Success % change:

$$\text{Net \% ch} = a + b + \frac{ab}{100}$$

$$a = 25\%, \quad b = 20\%$$

Cement

$$25 + 20 + \frac{25 \times 20}{100}$$

$$= 50\%$$

$$\text{Soda} = 25 + 10 + \frac{25 \times 10}{100}$$

$$= 37.5\%$$

$$1998(120) \quad 2000(150)$$

25% ↑

Cement

15%

18%

20% ↑

Soda

20%

22%

10% ↑



## QUESTION-1



increase.

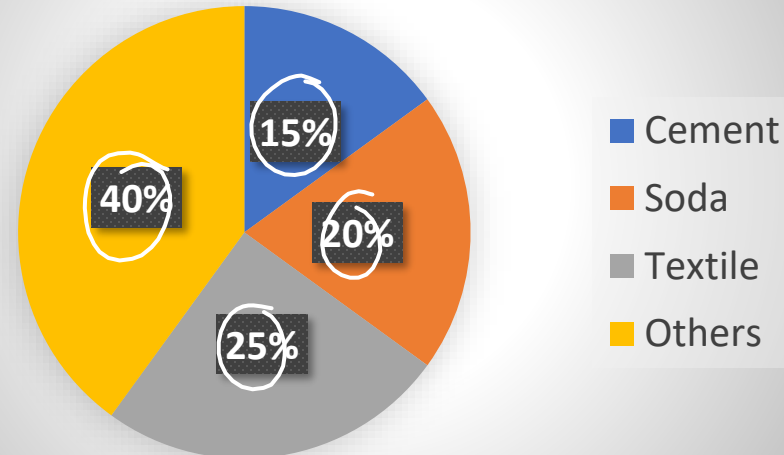
#Q. Which product segment has registered the maximum percentage growth in revenue during a two-year period?

- ✓ A. Textile → 40% ↑
- B. Cement → 20% ↑
- C. Soda → 10% ↑
- D. Others X

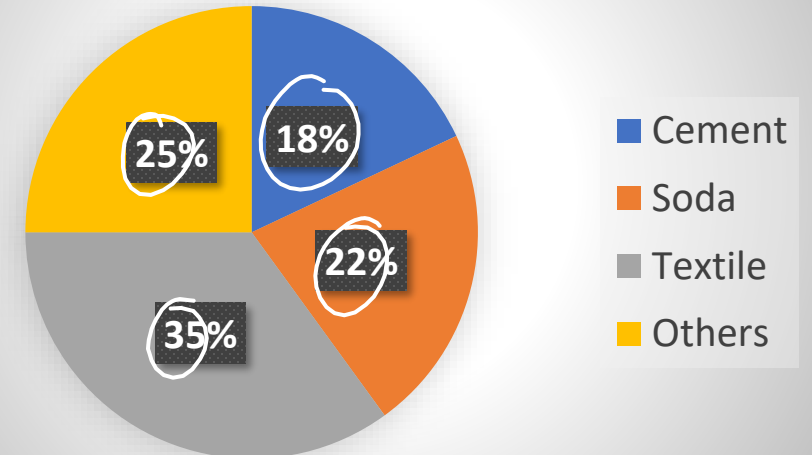
$$a + b + \frac{ab}{100}$$

a =

Sales for Year 1998 = 120 Cr



Sales for Year 2000 = 150 Cr



## QUESTION- 2



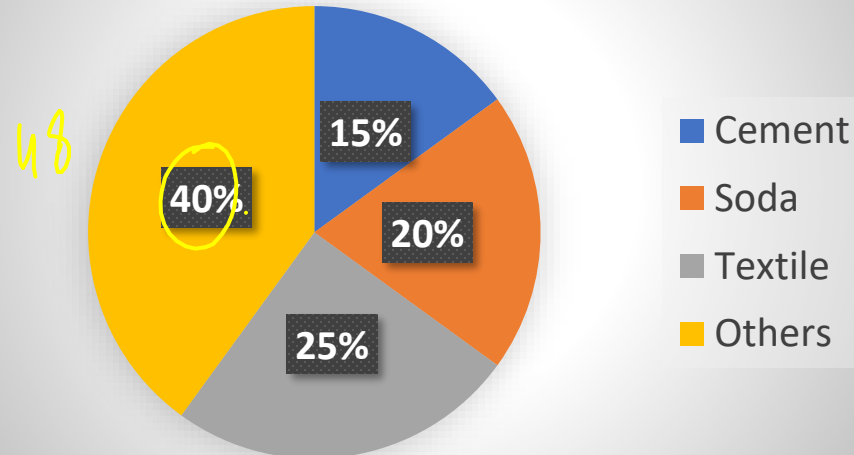
#Q. What is the annual percentage growth in revenue of the 'Others' segment?

$$\left. \begin{array}{l} a = 25\% \\ b = \end{array} \right\} \text{Net \% ch} =$$

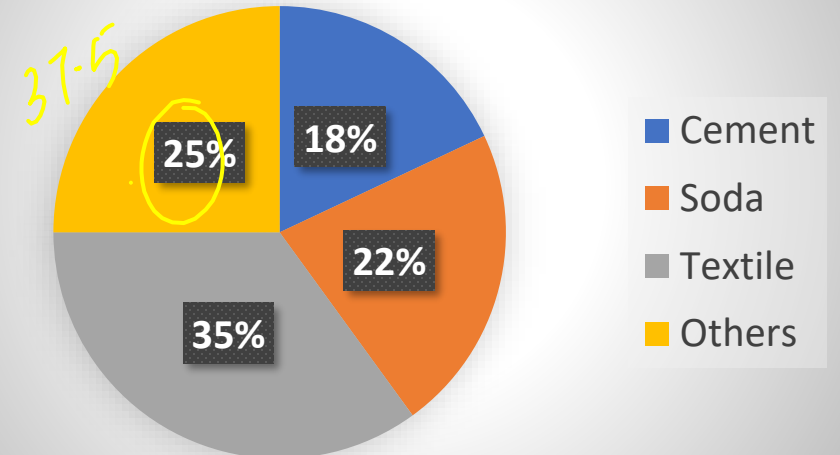
$$48 \rightarrow 37.5$$

$$\frac{21.875\%}{2} \downarrow =$$

Sales for Year 1998 = 120 Cr



Sales for Year 2000 = 150 Cr



### QUESTION- 3

$$\frac{3}{4} \times 52.5$$

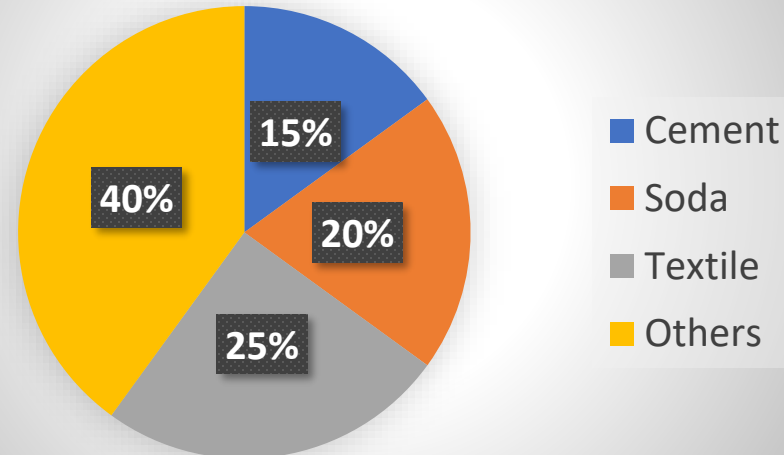


#Q. If the textile segment continues the current trend in growth rate, then how much sales will the company generate from textile in 2002?

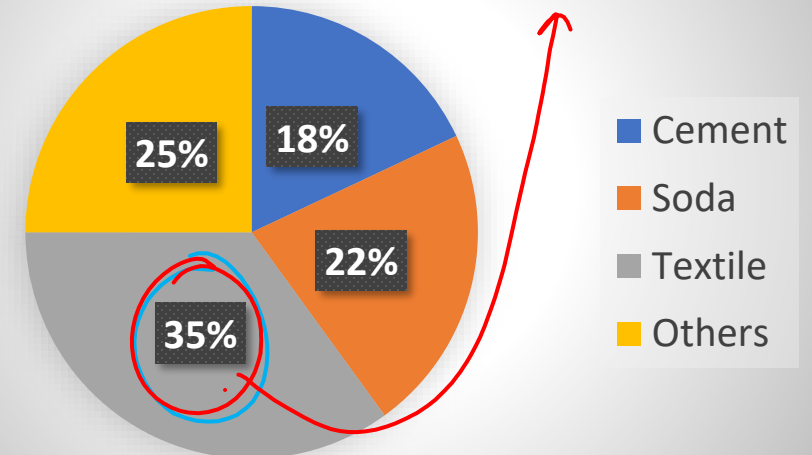
- A. 90 crore
- B. 91.875 crore
- C. 87.5 crore
- D. 85 crore



Sales for Year 1998 = 120 Cr



Sales for Year 2000 = 150 Cr



## QUESTION- 4



#Q. What is the annual <sup>Simple.</sup> percentage growth of revenue that the cement segment has registered for the company?

A. 50%

B. 45%

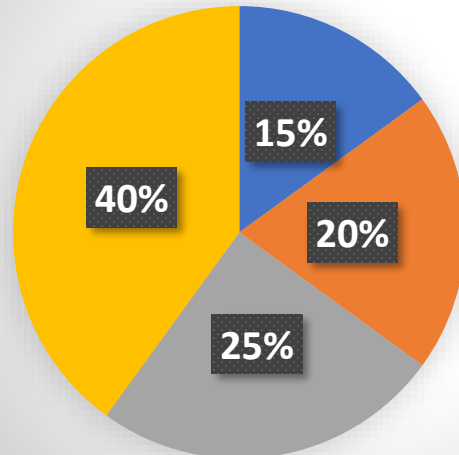
C. 55%

D. None of the above

Net % ch = 50%

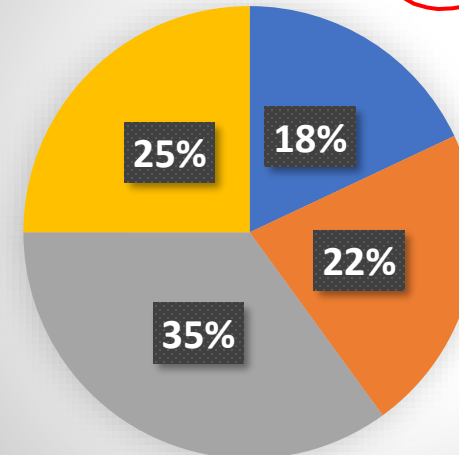
25%.

Sales for Year 1998 = 120 Cr



■ Cement  
■ Soda  
■ Textile  
■ Others

Sales for Year 2000 = 150 Cr



■ Cement  
■ Soda  
■ Textile  
■ Others



**SET - 3**



Four sisters Suvarna, Tara, Uma, and Vibha are playing a game such that the loser doubles the money of each of the other players. They played four games and each sister lost one game in alphabetical order (increasing). At the end of the fourth game, each sister had Rs 32.

Q : Who started with the minimum amount?

Q : What was the amount with Tara at the beginning?

Q : What was the amount with Vibha at the beginning?



Four sisters Suvarna, Tara, Uma, and Vibha are playing a game such that the loser doubles the money of each of the other players. They played four games and each sister lost one game in alphabetical order (increasing). At the end of the fourth game, each sister had Rs 32.





## SET - 4

Read the following data and answer the given questions.



An ATM dispenses exactly Rs. 5000 per withdrawal using notes of Rs. 100, Rs. 200, Rs. 500.

The ATM requires every customer to give her preference for one of the three denominations of notes.

It then dispenses notes such that the number of notes of the customer's preferred denomination exceeds the total number of notes of other denominations dispensed to her. <sup>to</sup>

|     |    |   |   |   |
|-----|----|---|---|---|
| 100 | 0  | 1 | 4 | 6 |
| 200 | 0  | 2 | 3 | 2 |
| 500 | 10 | 9 | 8 | 8 |

## QUESTION-1



7

#Q. In how many different ways can the ATM serve a customer who gives Rs. 500 notes as her preference?

|     | 1  | 2  | 3  | 4  | 5  | 6 | 7 | 8  | 9  | 10 |                 |
|-----|----|----|----|----|----|---|---|----|----|----|-----------------|
| 100 | 0  | 5  | 3  | 1  | 10 | 8 | 6 | 4  | 2  | 0  | 15 13           |
| 200 | 0  | 0  | 1  | 2  | 0  | 1 | 2 | 3  | 4  | 5  | 0 1 2 3 4 5 6 7 |
| 500 | 10 | 9  |    |    | 8  |   |   | 8  |    |    | 7               |
|     | 10 | 14 | 13 | 12 | X  |   |   | 15 | 14 | 13 |                 |

An ATM dispenses exactly Rs. 5000 per withdrawal using notes of Rs. 100, Rs. 200, Rs. 500.

The ATM requires every customer to give her preference for one of the three denominations of notes.

It then dispenses notes such that the number of notes of the customer's preferred denomination exceeds the total number of notes of other denominations dispensed to her.

## QUESTION- 2



#Q. If the ATM could serve only 10 customers with a stock of fifty 500-rupee notes and a sufficient number of notes of other denominations, what is the maximum number of customers among these 10 who could have given 500-rupee notes as their preferences?

10 notes each

$$50/10 = 5$$

↑ notes

$$50/9 = 5$$

$$50/8 = \textcircled{6} \checkmark$$

An ATM dispenses exactly Rs. 5000 per withdrawal using notes of Rs. 100, Rs. 200, Rs. 500.

The ATM requires every customer to give her preference for one of the three denominations of notes.

It then dispenses notes such that the number of notes of the customer's preferred denomination exceeds the total number of notes of other denominations dispensed to her.

### QUESTION- 3



$$50 \times 10 = 500$$

$$10/9/8$$

#Q. What is the number of 500-rupee notes required to serve 50 customers with 500-rupee notes as their preferences and another 50 customers with 100-rupee notes as their preferences, if the total number of notes to be dispensed is the smallest possible?

$$50 \times 8 = 400$$

A. 750

B. 800

C. 1400

D. 900

|     |   |   |  |    |
|-----|---|---|--|----|
| 100 | → | 5 |  | 10 |
| 200 | → |   |  |    |
| 500 | → | 9 |  | 8  |

An ATM dispenses exactly Rs. 5000 per withdrawal using notes of Rs. 100, Rs. 200, Rs. 500.

The ATM requires every customer to give her preference for one of the three denominations of notes.

It then dispenses notes such that the number of notes of the customer's preferred denomination exceeds the total number of notes of other denominations dispensed to her.

## QUESTION- 4



#Q. What is the <sup>min.</sup>maximum number of customers that the ATM can serve with a stock of fifty 500-rupee notes and a sufficient number of notes of other denominations, if all the customers are to be served with at most 20 notes per withdrawal?

A. 16

B. 10

C. 12

D. 13

|          |    |    |    |    |
|----------|----|----|----|----|
| 100 →    | 1  |    | 1  | 0  |
| 200 →    | 22 | 20 | 17 | 15 |
| 500 →    | 1  | 2  | 3  | 4  |
| 50 notes | X  | X  | X  | 19 |

$$\frac{50}{4} \rightarrow 12$$

An ATM dispenses exactly Rs. 5000 per withdrawal using notes of Rs. 100, Rs. 200, Rs. 500.

The ATM requires every customer to give her preference for one of the three denominations of notes.

It then dispenses notes such that the number of notes of the customer's preferred denomination exceeds the total number of notes of other denominations dispensed to her.





THANK  
*You*

